

SpectroCloud × VAST — Engagement Record

A record of the work delivered against `SCOPING.md` and `VALIDATION-PLAN-RESPONSE.md`.

Substrate: VAST-on-Cloud (VoC) on AWS · region `us-east-2` · dedicated VAST VPC `10.20.0.0/16` · (VMS `10.20.4.81`, ENode `i3en.24xlarge`) · Palette-managed EC2 Kubernetes clusters (SaaS `console.spectrocloud.com`, project *ISC-Strategic-Alliance*, and self-hosted).

This record tracks the two scoping documents section-by-section. Status legend: 

Delivered ·  **Validated** (delivered *and* exercised end-to-end with a passing test) · **Partial** · ☐ **Deferred / next increment.**

1. Executive summary

The scoping documents defined a joint architecture to stand up **VAST-on-Cloud on AWS** and consume it from **Palette-managed Kubernetes** via the VAST **CSI / COSI / block** drivers, with **multi-tenant isolation** and **QoS**. That architecture is now **built, automated, and validated end-to-end**:

- **VoC deployed** on AWS via the Polaris / `vastcloud` path, in its **own VPC** (the recommended **Option B** topology), with a codified **deploy** → **tenant** → **teardown lifecycle**.
- **Network foundation** (VPC, peering, security groups, tenant client-IP ranges) delivered as **Terraform**, peering the VAST VPC to **four** consuming cluster VPCs.
- **Three VAST packs** (`vast-csi` , `vast-cosi` , `vast-block` , chart `2.6.5` , rev `-4`) built, published to **OCI/ECR** registries, and wired into a **Palette cluster profile** with VMS connection details as per-cluster sensitive variables.
- **Multi-tenancy proven**: isolated VAST tenants (VIP pool + view + view policy + QoS + client-IP range) per Palette cluster.
- **Storage validated**: **NFS RWX** `PASS` and **NVMe/TCP block** `PASS` end-to-end against isolated tenants, with a **self-service verification kit**.

The result covers **Validation-Plan Phase 1 (CSI)** and **Phase 2 (multi-tenancy)** in full; **Phase 3 (PaletteAI / DataEngine / DataBase)** remains the next increment, gated on VAST-supplied services.

2. SCOPING.md — build mechanics, section by section

§	Scoped item	Status	Delivered as
1	New Palette-managed EC2 K8s cluster consuming VAST (CSI/NFS primary; S3/block secondary)		Multiple Palette clusters consuming VAST over CSI (NFS), COSI (S3), block (NVMe/TCP)
3	VoC via Polaris + <code>vastcloud</code> (6-stage preflight)	✓	Deployed; <code>voc/scripts/polaris-register.sh</code> , <code>deploy-voc.sh</code> wrap the flow; VMS live at <code>10.20.4.81</code>
4	CSI controller → VMS + node → VIP reachability; multi-tenancy by Client IP Range		Both reachability paths validated; per-tenant client-IP ranges enforced (<code>tenants.json</code>)
5	Network topology decision — Option B (VAST in own VPC, peered)	✓	Realized: VAST VPC <code>10.20.0.0/16</code> peered to cluster VPCs <code>10.30 / 10.40 / 10.50</code> + in-VPC
6	Capacity & instance sizing	✓	VoC on <code>i3en.24xlarge</code> ; cluster CP <code>m5.xlarge</code> (≥4 vCPU floor confirmed), workers <code>m5.xlarge</code>
7	Phased execution plan (Phase 0–5)		See §4 below — Phases 0–4 delivered/validated, Phase 5 partial
10	Terraform owns the connective substrate (plan → review → apply)	✓	<code>voc/terraform/*</code> (network, peering, tenancy), <code>spectro-peering/</code> , <code>voc-tenants/</code>
11	Build the <code>vast-csi</code> Palette pack (chart 2.6.5)		Built + published + attached + validated (extended to COSI + block too)
12	Firewall / ports on SGs	✓	SG rules for NFS 2049/111/20048(+20106-8), VMS 443/80, block 4420/4520/8009, RPC, SSH
13	RDMA/EFA finding — no RDMA path to VAST on AWS; plan high-bandwidth TCP	✓	Confirmed and planned around (NVMe/TCP + NFS over TCP data path; EFA scoped as optional GPU-only add-on)

3. VALIDATION-PLAN-RESPONSE.md — validation results

Phase 1 — VAST CSI on Palette — Validated

- **Packs built & published:** `vast-csi` , `vast-cosi` , `vast-block` (chart `2.6.5` , revision `-4`) → OCI/ECR (**ISC Palette Registry**, both SaaS and self-hosted).
- **Cloud CSI tuning** from the scope applied in pack values (TCP data path, VIP-pool selection, mount options).
- **Result:** **NFS VERIFY: PASS** (RWX provision + read/write) and **NVME/TCP VERIFY: PASS** (block RWO) on live Palette clusters; all driver pods (`vast-csi` , `vast-cosi` , `vast-block`) Running; StorageClasses `vastdata-filessystem` + `vastdata-block` present.

Phase 2 — Multi-tenancy architecture — Validated

- **Isolated tenants** provisioned per consuming cluster, each = **tenant + VIP pool + view policy + view + QoS tier**, gated by **client IP range** (`voc/scripts/tenant-voc.sh` , `tenants.json` ; idempotent, one entry per cluster).
- **QoS tiers** defined (bronze / silver / gold — BW + IOPS Max/Burst).
- **Palette → VAST mapping** realized: each cluster → its own tenant → StorageClass (VIP pool + path baked in); VMS credentials/endpoint delivered as **per-cluster sensitive profile variables**, never baked into the published profile.
- **Result:** separate clusters (`fieldeng` , `spectro` , `voc-tenant-1` @ `10.40/16` , `voc-tenant-2` @ `10.50/16`) each reach **only** their own VIP pool / view; validated independently.

Phase 3 — PaletteAI Profile Bundles (DataEngine / DataBase) — ☐ Deferred

- Next increment; gated on VAST-supplied DataEngine-enabled cluster + GA confirmation (see `VALIDATION-PLAN-RESPONSE.md` §E). Not started by design.

4. Phased execution plan (`SCOPING.md` §7) — completion

Phase	Status	Notes
0 — Decisions & procurement	✓	BYOL/Polaris obtained; Option B topology chosen; Palette-into-VPC behavior confirmed
1 — Network foundation	✓	VAST VPC + subnets + IGW/NAT; peering to 4 cluster VPCs with return routes; SGs; tenant client-IP ranges — all Terraform
2 — Deploy VAST (VoC)	✓	Polaris template + <code>vastcloud create</code> ; post-deploy tenants/VIP-pools/views/policies automated; VMS endpoint persisted to SSM
3 — Provision Palette clusters		Cluster profile <code>vast-storage-aws</code> (OS+k8s+CNI+CSI + 3 VAST packs); clusters deployed on SaaS + self-hosted; node → VIP + controller → VMS reachability verified
4 — Build vast-csi pack & validate		Packs built/published/attached; privileged namespaces handled; <code>vast-mgmt</code> secret from profile vars; RWX PVC + block PVC validated
5 — Hardening & ops	▸	Multi-tenant isolation + QoS delivered; lifecycle automation + teardown in place; remaining: per-AZ NAT HA review, monitoring dashboards (Prometheus/Grafana), documented runbook

5. Deliverables inventory

Infrastructure-as-Code (Terraform) - `voc/terraform/` — VAST VPC network, VoC lifecycle substrate, VMS endpoint outputs - `voc/terraform/spectro-peering/` — cross-account peering (spectro ↔ VAST VPC) - `voc/terraform/voc-tenants/` — reusable per-tenant peer-VPC module (workspace-per-tenant) -

`terraform/network/` , `terraform/vast-tenancy/` — network + VAST tenancy modules

Automation (`voc/scripts/`) - `polaris-register.sh` · `deploy-voc.sh` · `destroy-voc.sh` — VoC lifecycle - `tenant-voc.sh` — idempotent VMS multi-tenancy setup (tenant/VIP-pool/view/policy/QoS) - `validate-voc.sh` — post-deploy validation + VMS endpoint persistence - Driven by a CodeBuild lifecycle pipeline (deploy / tenant / teardown actions)

Go tooling (`voc/pack_devel/`) - `packsync` — production-grade pack publisher (ECR/OCI + Palette sync; 96%+ test coverage, CI, Dockerized builder) - `vast-profile` — `create-profile` (builds+publishes the cluster profile, resolves pack UIDs by name) + `deploy-cluster` (binds a cluster, fills VMS vars per-tenant) - `packtest` — pack validation harness

VAST packs — `vast-csi` , `vast-cosi` , `vast-block` (chart `2.6.5` , revisions `-1` ... `-6` ; `-4` published)

Verification kit (`voc/tenant-verification/`) — self-service NFS / NVMe-TCP-block / COSI-S3 Jobs (`01` / `02` / `03`) + README; each writes-and-reads a token and prints `VERIFY: PASS`

Documents — `SCOPING.md` , `VALIDATION-PLAN-RESPONSE.md` , `PACKS.md` , `EMAIL-TO-VAST.md` , `voc/README.md` , this record

6. Validation evidence (as run)

Test	Driver / class	Result
NFS filesystem, RWX	<code>csi.vastdata.com</code> / <code>vastdata-filestorage</code>	NFS VERIFY: PASS — PVC Bound (VMS-provisioned), read/write verified
NVMe/TCP block, RWO	<code>block.csi.vastdata.com</code> / <code>vastdata-block</code>	NVME/TCP VERIFY: PASS — PVC Bound, device attached + mounted, read/write verified
COSI / S3 object	<code>objectstorage.k8s.io</code> (COSI)	Drivers Running; provisioner validated; bucket test available (per-tenant edits)
Tenant isolation	VIP pool + view + client-IP range	Proven — each cluster reaches only its own tenant's pool/view
Control-plane reach	CSI controller → VMS <code>:443</code>	Verified (volumes provision on demand → VMS API reachable + authorized)
Data-path reach	node → VIP pool (NFS/block)	Verified (mounts succeed over the peered path)

7. Deviations from scope & notable findings

- **Test substrate moved to a dedicated VAST VPC** (`10.20.0.0/16`) peered to multiple cluster VPCs — this *is* the recommended Option B, realized at wider scale (four peers) than the single-peer baseline in the scope.

- **Scope extended beyond NFS-only:** the scope's primary was NFS CSI; delivered all three — **NFS + COSI/S3 + NVMe/TCP block** — as separate validated packs.
 - **Control-plane automation:** the scope proposed the `vast-data/vastdata` Terraform provider as the cleanest path; delivered via an **idempotent imperative VMS-API flow** (`tenant-voc.sh`) that maps 1:1 to the same objects — interchangeable with the TF-provider path.
 - **Instance-sizing finding:** Palette enforces a **≥4 vCPU floor per control-plane pool** — a single `m5.large` (2 vCPU) CP is rejected; `m5.xlarge` is the practical minimum.
 - **Block mount `blkid` settle:** on first mount a fresh NVMe/TCP device can hit a transient `blkid` probe timeout that self-resolves on retry — node-side, not a VAST fault (confirmed by a clean `PASS` on retry / on other nodes).
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8. Open items (next increment)

1. **Phase 3 / PaletteAI** — DataEngine + DataBase bundles, pending a VAST-supplied DataEngine-enabled cluster + GA confirmation.
2. **Ops hardening (Phase 5)** — per-AZ NAT for HA, Prometheus/Grafana per-tenant dashboards, documented runbook.
3. **Pack certification** — a *validated* (vs community) VAST pack via the Spectro Cloud Partner Program.
4. **Consolidated asks to VAST** — see `VALIDATION-PLAN-RESPONSE.md` §E (remaining: DataEngine GA, sample-app repos, certification path).